

Education Algorithm — Enterprise Master Migration Blueprint

Strict Zero-Bug Architecture: Java 21 LTS + Spring Boot 3.3 + Spring Security 6 + React 18 SPA (Planning Only)

1. Executive Summary & Architectural Governance

This official Master Blueprint details the migration of Education Algorithm from PHP/Apache to a bank-grade **Java 21 Spring Boot 3.3** microservice/modular monolith backend paired with a **React 18 SPA** frontend. **Governance Rule:** This document is an architectural planning specification. No code execution, file mutation, or infrastructure deployment is to be performed prior to formal Phase Gate approvals.

2. Architecture Decision Records (ADR-001)

ADR-001 (Architectural Style Gate): Evaluating Option A (Modular Spring Boot Monolith - Recommended) vs Option B (Microservices) vs Option C (Hybrid Extraction). Gate 2 approval required prior to development.

3. 193-File Inventory Reconciliation

Explicit application category counts account for 111 core files (13 Public, 11 Enrollment, 25 LMS, 18 Admin, 19 Faculty, 15 APIs, 10 Includes). The remaining 82 workspace files are flagged as **REQUIRES SOURCE RECONCILIATION** and will be validated during Phase 1 pre-migration audit.

4. Functional Migration Matrix

Legacy Feature	Target Component / API	Security & Business Rules
Course Catalog (courses.php)	CoursesPage.jsx GET /api/v1/courses	Public access, published cohort filtering.
Email OTP Dispatch (api-enrollment.php)	EnrollmentService.java POST /api/v1/enrollment/send-otp	SHA-256 IP/Email rate limiting, 6-digit OTP, 10-min expiry.
Coupon Math & Rounding	CouponService.java POST /api/v1/enrollment/validate-coupon	Percentage/fixed math, BigDecimal HALF_UP, integer paise conversion.
Razorpay Checkout Order	RazorpayService.java POST /api/v1/enrollment/create-order	Auth Bearer token, exact paise calculation, amount discrepancy defense.
Razorpay Webhook (webhook-razorpay.php)	RazorpayWebhookController.java POST /api/v1/payments/razorpay-webhook	HMAC-SHA256 signature verification, idempotent transaction fulfillment.
Student LMS (dashboard.php)	StudentDashboardPage.jsx GET /api/v1/student/dashboard	ROLE_STUDENT RBAC, progress metrics, DRM video streaming.
Admin Finance (admin-finance.php)	AdminFinanceStudioPage.jsx GET /api/v1/admin/finance	ROLE_ADMIN RBAC, revenue aggregations, coupon management.

5. Zero-Defect Exception Matrix

Exception Class	HTTP Code	Response Contract
MethodArgumentNotValidException	400	{"success": false, "code": "VALIDATION_FAILED"}
CouponNotFoundException	400	{"success": false, "code": "INVALID_COUPON"}
OtpInvalidException / OtpExpiredException	400	{"success": false, "code": "INVALID_OTP"}
UnauthorizedException	401	{"success": false, "code": "UNAUTHORIZED"}
AccessDeniedException	403	{"success": false, "code": "FORBIDDEN"}
RateLimitExceededException	429	{"success": false, "code": "RATE_LIMITED"}

DatabaseException	500	{"success": false, "code": "INTERNAL_ERROR"}
-------------------	-----	--

6. 9-Phase Cutover Plan & Governance Gates

Cutover Phasing: Phase 1 Build → Phase 2 Test → Phase 3 Data Rehearsal → Phase 4 Parallel Validation → Phase 5 Canary (5% traffic) → Phase 6 Monitoring → Phase 7 Full DNS Cutover → Phase 8 Standby Window (72h) → Phase 9 Retirement.

Governance Gates: 11 mandatory sign-off gates (Gates 1-11) enforce strict architectural, financial, and security compliance.